

CHECKLIST

SMALL CRAFT - ELECTRICAL/ELECTRONIC CONTROL SYSTEMS FOR STEERING, SHIFT AND THROTTLE

Ref.: EN ISO 25197:2018 (ISO 25197 2012+A1:2014)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



Subject to check	Clause	Requirements	Checked ?
1 Label: Operational characteristics, instructions and warnings for proper use. Affixed: on-product.	4.11	[Yes / NA ?]	
2 Label: Portable helm station controls with following information: -keep proper lookout; -hold on to prevent falling, boat may move suddenly; -read owner's manual for safe use of the system. Affixed: on the portable helm or where the device is stored.	7.3	[Yes / NA ?]	
3 Label: Warning labels or text for the DPS and/or if helm stations not equipped with a display screen: -boat is considered underway; -keep proper lookout; -stay out of water; -hold on to prevent falling, boat may move suddenly; -read owner's manual for safe use of the system.	8.3	[Yes / NA ?]	
4 Label: If the helm station(s) is (are) not equipped with an electronic display and the craft has a dynamic-positioning system installed. Use symbols or the following warnings: -boat is considered underway; -keep proper lookout; -propellers are spinning, swimming is dangerous; -boat may move suddenly, hold on to prevent falling; -read owner's manual for safe use of the system. Affixed: in the vicinity of the helm station.	11	[Yes / NA ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

Subject to check	Clause	Requirements	Checked ?
5 The manufacturer of the system has submitted a Declaration of Conformity with regards to the compliance as Annex II component according to the Recreational Craft Directive for the parts of the system intended for steering control of the boat. See also comment on last page of this checklist.		[Yes / NA ?]	
6 The manufacturer of the system has submitted a confirmation with regards to the compliance to ISO 25197 for the parts of the system intended for throttle control of the boat. See also comment on last page of this checklist.		[Yes / NA ?]	

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7	The system is energized whenever the propulsion engine(s) are running.	4.4	[Yes ?]	
8	The system is fully operational within five seconds after being powered (except for dynamic positioning and displays).	4.5	[Yes ?]	
9	Multi-installed engine steering systems are redundant by virtue of the fact that they are both mechanically and electrically independent of each other.	4.6	[Yes / NA ?]	
10	A single device is allowed for control of multiple engines (e.g. steering wheel, joystick).	4.6	[Yes / NA ?]	
11	Each helm station shall give a visual indication when active.	4.7	[Yes ?]	
12	When the system enters a fail-safe mode, the operator is alerted by a visible and/or audible means at each helm station.	4.8	[Yes ?]	
13	The sound pressure of an audible alarm 1 metre from the command station is at least 75 dB(A), but not greater than 85 dB(A). Systems incorporating a mute feature shall maintain the visual alert as long as the failure persists.	4.9	[Yes ?]	
14	Muting of the alarm is not allowed if the alert system is only audible.	4.9	[Yes ?]	
15	Instructions for proper installation and use of the steering system shall be made available by the manufacturer.	4.10	[Yes ?]	
16	It is only be possible to start propulsion equipment in neutral (exception: temporary override for emergency situations).	4.12	[Yes ?]	
17	Within 0,5 seconds on a physical input command, the steering, shift and throttle actuators do react/adjust to the input.	4.13	[Yes ?]	
18	ISO 8846 is meet for electrical components intended to be installed in petrol engine or petrol tank compartments for IGP.	4.16	[Yes ?]	
19	If the system provides both, cruising- and manoeuvring mode, an indication to the operator at the command station is provided of which mode the system is in. The system does not change the mode without input from the operator.	4.18	[Yes / NA ?]	
20	If the system is equipped with a control head, requirements of clause 5 apply. [operator input device, other than a steering wheel, for the simultaneous control of steering and propulsion]	5		
21	The control head operation is permitted for both cruising-mode and manoeuvring-mode operation.	5.1	[Yes / NA ?]	
22	When the operator releases the grip, the control head position shall return to the neutral.	5.2	[Yes / NA ?]	
23	When released in cruising mode, the control head engine throttle control must not be returned to a low RPM or a manufacturer-determined idle state for operation.	5.3	[Yes / NA ?]	
24	Releasing the control head in manoeuvring mode to neutral results in a disengaged transmission and determined idle or stop state of electric motors.	5.4	[Yes / NA ?]	
25	The control head orientation relative to the craft and the movement of the craft are identically.	5.5	[Yes / NA ?]	
26	The orientation relative to the craft is clearly indicated for portable helms.	5.6	[Yes / NA ?]	
27	If the control head includes a rotation function, the control head activation, clockwise or counter-clockwise, results in rotating the craft in the same direction.	5.7	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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28	Transfer of command from one station to another is completed at the helm station intended to be active.	6.0	[Yes / NA ?]	
29	Portable helm station controls can only be used in the manoeuvring mode.	7.1	[Yes / NA ?]	
30	In case that a portable helm station control has of a loss of communication or malfunction it results in disengaged transmission and idling thermal engine(s) and stopping electric engine(s). In addition the operator is notified and able to transfer the function to another helm.	7.2	[Yes / NA ?]	
31	The signal strength of a wireless device is displayed at the portable control or an audible warning signal alerts the operator that the signal is weak and he/she is about to lose control.	7.4	[Yes / NA ?]	
32	A wireless device shall only be able to control the boat of origin.	7.5	[Yes / NA ?]	
33	An indication of its electrical-charge status is given on a wireless portable helm control and the device shall prevent activation when the charge is insufficient to maintain a connection for 15 minutes without loss of the wireless communication link.	7.6	[Yes / NA ?]	
34	A wireless portable helm control notifies the operator when the device is 15 minutes from deactivation due to insufficient charge.	7.7	[Yes / NA ?]	
35	Check point 8.1 to 8.8 only if a Dynamic Positioning System (DPS) is part of the system. Otherwise scratch out 8.1 to 8.8.	8.0		
36	A Dynamic Positioning System (= DPS) can only be activated manually.	8.1	[Yes / NA ?]	
37	The craft main helm station has a visual display for the DPS precision value.	8.2	[Yes / NA ?]	
38	Activation of the DPS is only be possible if the DPS precision value is within the manufacturer-set limits.	8.4	[Yes / NA ?]	
39	The DPS control system alerts the operator visually and audibly about disengagement in case that the DPS precision value of an activated dynamic-positioning (autonomous) mode is out of the manufacturer-set limits.	8.5	[Yes / NA ?]	
40	The manufacturer has set the maximum allowable envelope/radius and heading deviation. This one is not adjustable by the operator.	8.6	[Yes / NA ?]	
41	The manufacturer has set the maximum engine speed for dynamic positioning. This one is not adjustable by the operator.	8.7	[Yes / NA ?]	
42	In case that a command station in a multiple helm system has a malfunction, the system does not prevent transfer or operation from other helm stations. In addition the operator is notified audible and/or visual about the malfunction.	9.1.1	[Yes / NA ?]	
43	In the event of loss of steering control affecting only one engine in a multi-engine installation, the system is still be capable of steering the boat.	9.1.2	[Yes / NA ?]	
44	In the event of a command station failure, the operator shall be notified and the affected command station is switched to a fail-safe mode.	9.1.3	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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45	In the event of loss of steering control in a single-engine rudder or strut installation, emergency control of the rudder is possible.	9.1.4	[Yes / NA ?]	
46	The system notifies the operator of a command logic loss or a malfunction in its computer command logic.	9.2	[Yes ?]	

Instructions to be included in the owner's manual

46	Operational characteristics, instructions and warnings for proper use are described (if not labelled on-product).	4.11	[Yes / NA ?]	
47	Necessary information for safe operation of the system.	12	[Yes ?]	
48	Maintenance requirements.	12	[Yes ?]	
49	Locations of helm stations.	12	[Yes ?]	
50	Information regarding all visual indicators and audible alerts.	12	[Yes ?]	
51	Information on how to transfer command from one helm station to another.	12	[Yes ?]	
52	Warning symbols or text for the DPS systems: -do not leave helm station unattended; -boat is considered underway; -keep proper lookout; -stay out of water, propellers are spinning; -hold on to prevent falling, boat may move suddenly.	12	[Yes / NA ?]	

Note:

With regards to the certification of electrical/electronic control systems for steering, shift and throttle, following Recommendation For Use (RFU) has been published by the Recreational Craft Sectorial Group (RSG) as RFU #115:

Scenario/Question:

Are shift and throttle and dynamic position control systems, or combinations thereof, which are included in scope of standard ISO 25197, Annex II components?

Recommended Solution:

No. Only those components that control the steering that are listed in Annex II.3: steering wheels, steering mechanisms and cable assemblies.

A "control head" (e.g. a joystick) which controls both the throttle and steering systems are considered to be Annex II components but a throttle of a shift, which is independent of the system that controls the attitude of the drive unit/rudder, is not.

Comments:
